

DAY 03 · THE AI DEV TOOLCHAIN

One prompt, five hops.

Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

1. Home screen

> Explain APIs like I'm 12

Run ▶

01

User

a person with a goal

→

02

Prompt

the goal, as text

→

03

LLM

the model thinks

→

04

API

the bridge, both ways

→

05

Application

the answer, delivered

● ● ● trace.log

```
# press Run to trace a request through the pipeline –  
or click any stage below for details
```

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2. Enter prompt
(optional)

> Explain APIs like I'm 12

Run ▶

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One prompt, five hops.

Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

> Generate me 5 topics about machine learning

Run ▶

3. Click run

01

User

a person with a goal



02

Prompt

the goal, as text



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LLM

the model thinks



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4. Watch the
pipeline stream

01

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a person with a goal



02

Prompt

the goal, as text



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the model thinks



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the bridge, both ways



05

Application

the answer, delivered

● ● ● trace.log

> run

[01 user] intent captured — someone wants an answer

[02 prompt] "Generate me 5 topics about machine learning" packed into a request

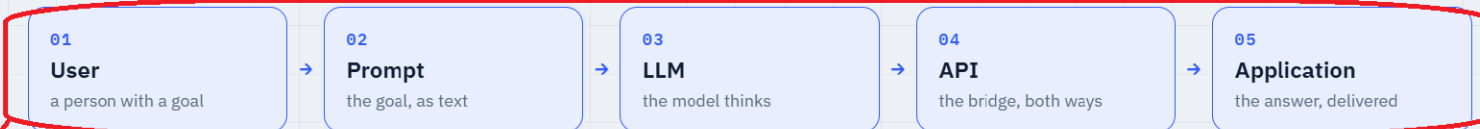
[03 llm] tokens in → tokens out · model is thinking...

[04 api] answer returned as JSON · 200 OK

[05 app] rendered on screen for the user

✓ pipeline complete in 1.02s

5. Click any stage to show more details about it



● ● ● trace.log

```
> run
[01 user]    intent captured — someone wants an answer
[02 prompt]  "Generate me 5 topics about machine learning" packed into a request
[03 llm]     tokens in → tokens out · model is thinking...
[04 api]     answer returned as JSON · 200 OK
[05 app]     rendered on screen for the user
✓ pipeline complete in 1.02s
```

STAGE DETAIL — CLICK A STAGE ABOVE

02 • Prompt

WHAT HAPPENS HERE

The goal becomes text the model can work with: the user's words, plus (often) hidden instructions the app adds behind the scenes — tone, format, extra context.

REAL EXAMPLE

In ChatGPT, your message travels together with a system prompt you never see.

TOOLS THAT LIVE HERE

When the prompt is part of a program, it's written in the IDE — a prompt is just a string until the pipeline moves it.

6. Detailed descriptions of the three main AI Development Toolchain

TASK 01 · THE TOOLS, IN MY OWN WORDS

Three tools, one job each

INTERFACE

API

A waiter between my code and someone else's service. I never walk into their kitchen — their servers, their model — I just hand a request to the waiter and it comes back with the answer. The menu is fixed: the API defines exactly what I can ask for and how.

```
my code → request → their service  
→ JSON → my code
```

Where it fits: inside the pipeline itself — it's the bridge between my Application and the LLM. Requests go in, answers come back.

TERMINAL

CLI

Talking to the computer with text instead of clicks. One line in the terminal installs a library, runs a script, or pushes code — no hunting through menus. Slower to learn, much faster to use.

```
pip install requests · git push ·  
curl api.model.com
```

Where it fits: around the pipeline, not inside it — it's how I install SDKs, run my script, and test API calls while building.

WORKSHOP

IDE

The workshop where the code gets written. Editor, autocomplete, debugger and a terminal, all in one app — so I'm not juggling ten separate tools while I build.

```
VS Code, PyCharm — mine is VS Code
```

Where it fits: where the Application step is born. The prompt strings and the API calls are all typed here first.

Spot the pipeline in the wild

ChatGPT

A website using AI

A Python script

7. View different scenarios of the same pipeline

User

You, in the browser

Prompt

Your message (+ hidden system instructions)

LLM

A GPT model in OpenAI's data center

API

OpenAI's own API connects the web app to the model

Application

The chat interface at chatgpt.com

It feels like one magic box, but it's the same five hops — OpenAI just built every one of them for you. You only ever touch step 01.